Report

On

“ Student Management ”

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[Year: 2025-26]

CERTIFICATE

This is to certify that **Dhage Vaishnavi (Roll No.:-33), Shrimangale Shivkumar(Roll No.:-34), Kazi Jiya Mukim (Roll No.:-35) , Shelke Vaishnavi (Roll No.:-36), Balapure Swati (Roll No.:-37) ,Somvanshi Krishnanand (Roll No.:-38)** has successfully completed their project on Student Management as a part of the academics for third year of the degree in computer science and Engineering as Prescribed by Dr. Babasaheb Ambedkar Technological University, Lonere during academic year 2025-26.

Prof. S.S.Gujar	**Prof.Dr. N. M. Halgare**	**Prof. Dr. S. M. Jagade**
Guide	**H.O.D**	**Principal**

ACKNOWLEDGEMENT

With immense pleasure, I am presenting the project Report as part of the curriculum of the third year Computer Science and Engineering. I wish to thank all the people who gave me an unending support from beginning of the project **.**

I express my sincere and profound thanks to **Prof. S.S.Gujar** our Project Guide and our HOD Prof .Dr. N. M. Halgare who always stood by us as the helping and guiding support and all those who have directly or indirectly guided and helped me in the preparation of the Project.

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ABSTRACT

In today's world the management system is very important and essential for every system. This management system is an application-based system, having two applications developed, one for teachers to manage teacher details and another for students to mark their details . Every organization whether government or private uses an information system to store data of their staff. However, in India it is found that many small-scale industries or colleges use pen and paper to keep a record. However, there are many advanced technology systems available that can do this work but they all are costly for these low-level industries.

This project is useful for easy user interface. The system uses the powerful database management system, data retrieval and data manipulation. This project provides more ease for managing the data than manually maintaining the data. Hence it saves the lot of time of ours also. So, we can say that the project is useful for saving valuable time and reducing huge paper work.

INTRODUCTION

The student management system is an environment where all the process of the student in the institution is managed. It is done through the automated computerized method. Conventionally this system is done using papers, files and binders. This system saves the time of the student and of the administrator.

It includes process like registration of student details like roll no , name ,marks etc. This system reduces the cost and workforce required for this job. As the system is online the information is globally present to everyone. This makes the system easy to handle and feasible for finding the omission with updating at the same time. As for the existing system, they use to maintain their record manually which makes it vulnerable to security.

As the number of student increases in the institute manually managing the strength becomes a hectic job for the administrator. This computerized system stores all the data in the database which makes it easy to fetch and update whenever needed. With the development of technology, the rise of digitization and the rise of sort social networks, the sharing of information online kind of basically has generally literally become very pretty quiet in a fairly major way in a sort of big way. As a result, the definitely really entire online system really kind of has mostly definitely become very popular over the basically for all intents and purposes past years in a sort of generally big way, which literally fairly significant.

During each step, technology programs and tools attempt to assist the research process and prove that although technology increases the quantity of skills and literacy needed to complete research, it also increases the efficient of each step and effectiveness of the finished product. Today, the innovations and improvements of technology have produced several assistances that are very much useful and convenient to the research and development departments. These assistance may be in the forms of programs and software’s that are largely applied and used in researches.

OBJECTIVE

Certainly the actually main really goal of this project for all intents and purposes definitely is as follows:

The fairly goal of my project for all intents and purposes basically is very very simple but also important and really me really basically want to offer a particularly very simple entertainment or entertainment solution to the masses in a particularly important way, or so they for all intents and purposes thought. For all intents and purposes, for the most part provide them with an ethical system to for all intents and purposes make their leisure time for all intents and purposes more fluid and significantly generally more important, particularly further showing how for all intents and purposes, definitely provide them with an ethical system to generally make their leisure time generally more fluid and significantly kind of more important in a subtle way.

The project is totally built at administrative end and thus only the administrator is guaranteed access. Functionalities provided by Student Management System-:

- Simplifying and Streamlining all Tasks.
- Better communication.
- Easy access to all.
- Easy to manage the data in efficient manner.
- Complete tracking of all the students with their proper details in very efficient manner.
- Timetable can be managed by the teachers very easily.

TOOLS AND TECHNOLOGY USED

Front End-:

- We will be using Angular for Front End, which is a JavaScript web framework builds application using MVC methodology.

- Along with Angular we will be using Typescript and JavaScript for adding functionalities or to make the pages dynamic.

- We will be using HTML inside the components for modelling the skeleton of our web application.

- For styling purposes, we will using Tailwind CSS. Also we will be adding Bootstrap 5 in the application for making layouts

Back End-:

- We will using Java Script for backend which will be used to make connection between our application and database .

- For creating servers and making API we will be using Express which is library of js.

- For storing the data and records of students and teachers we will be using Google firebase and firestore database.

- Postman is used for API testing . It is an HTTP client that tests HTTP request, utilizing a graphical user Interface , through which we obtain different types of responses that need to be subsequently validated

SYSTEM DESIGN

The student information management system's design includes the creation of a home page that allows all students, staff, and other users to access the system. Every system user has 13 their own username and password. The login form on the home page allows a new user to register, or a current user to login to the system by entering their username and password.

Student-:

Because every college student plays such an essential role, the student is the centre of attention. Students can access college information, course details, subject details, faculty details, training and placement cell information, and exam section information, where course details include information about the branch of study, the college's academic calendar, year-by-year subject offerings by the branch, subject details include the syllabus of the subjects, information about the staff handling the subjects, and the subjects in which he is currently enrolled. The information about the companies, as well as the eligibility criteria for attending recruitment, are included in the placement specific.

Faculty-:

Each instructor gets a single file where they can keep track of their schedules, students, and classroom information. Administrators may access up-to-date information about teachers and their classrooms at any time thanks to that single database file. Teachers can fill out classroom reports and forms faster utilizing the interactive teacher database because it has all of the necessary information. The form is automatically filled out with the teacher's name and classroom information. Teachers need to do nothing more than fill in the blanks and click submit. OK. Reports and forms are saved to the teacher's file automatically. They can also look at the student's information to get a better idea. the student's performance, as well as increasing the student's efficiency. The personnel is also kept up to speed by the college on any essuie that rises

HARDWARE and SOFTWARE

1.Hardware used :-

The primary hardware used for coding, designing, and testing the website was a personal computer with the following specifications:

- **Processor:** Intel Core i5 (10th Generation) or equivalent AMD Ryzen pro- cessor.
- **Ram:** 8 GB DDR4, which allowed smooth multitasking between code ed- itors, browsers, and graphic design software.
- **Storage:** 512 GB SSD, providing fast read/write speeds to improve the efficiency of compiling, loading, and saving project files.
- **Display:** A 15.6-inch Full HD (1920x1080) monitor to clearly visualize design layouts, image details, and code simultaneously.
- **Operating System:** Windows 10 (64-bit), providing a stable environment for running IDEs, browsers, and other development utilities.

2. Software used :-

Use of several software tools that facilitated coding, designing, testing, and version control. These tools the use were essential to ensure a smooth development process and The development of the burger king website involved to deliver a high- quality, functional website.

- **Code editor : visual studio code :-**
- **Web browser : Google chrome and Mozzila Firefox :-**
- **MySQL**
- **Version control : Git and Github :-**

SECURITY AND FEATURE

- **Centralized Data Management:** Consolidates all student information (admissions, grades, fees) into a single, secure location, reducing data inconsistency and risk.
- **Role-Based Access Control:** Ensures only authorized personnel can view or edit specific data based on their role, a critical layer of security.
- **Data Encryption and Secure Storage:** Uses encryption to protect sensitive data from unauthorized access and cyber threats. Cloud-based systems should use secure configurations to prevent vulnerabilities.
- Secure Login Protocols: Involves features like multi-factor authentication (MFA) and single-sign-on (SSO) to prevent unauthorized access, even if a password is compromised.
- **Regular Backups:** Includes regular data backups to protect against data loss from hardware failure, cyberattacks, or other issues.

Functional features

- **Automation:** Automates administrative tasks like attendance tracking, fee collection, and grade entry to improve accuracy and efficiency.
- **Communication Tools:** Provides built-in messaging, email notifications, and mobile apps to improve communication between the school, students, and parents.
- **Performance Tracking:** Allows for tracking and reporting on academic performance, which aids in data-driven decision-making.
- **Customization and Scalability:** Allows the system to be customized to the school's unique processes and can be scaled to handle growth in student numbers and features.
- **Financial Management:** Securely manages financial transactions like tuition fees, scholarships, and other payments.
- **Document Management:** Enables secure storage and retrieval of essential documents like enrolment forms, certificates, and compliance records.

CONCLUSION AND REFERENCES

Conclusion: -

It's usually a good idea to go with a student information system that's built on a current system architecture to keep up with changing needs. This system should include wellorganized data coding and clearly defined business applications. The system's overview elucidates the convenience of exact data delivery at the tip of your fingertips, increasing student retention and teaching them how to manage their time effectively.

The proposed method is efficient and user-friendly, based on the results of the experiments and tests. In comparison to current methods of managing academic institutions, this project, which produces centralised software, makes work administration and management easier and gives full information about the issue of users' interest with just one mouse click. An easy-to-use user interface centralised software can be offered to the educational institution, allowing all services linked with the university to interact with one another and share data

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